Expno : 1

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SOLVING THE 8 QUEENS PROBLEM

PYTHON CODE :

```
N = 8

board = [[0] * N for _ in range(N)]

def is_safe(row, col):
    for i in range(row):
        if board[i][col] == 1:
            return False
        if col - (row - i) >= 0 and board[i][col - (row - i)] == 1:
            return False
        if col + (row - i) < N and board[i][col + (row - i)] == 1:
            return False
    return True

def solve(row):
    if row == N:
```

```
        for r in board:
            print(" ".join("Q" if x else "." for x in r))
        print()
        return

    for col in range(N):
        if is_safe(row, col):
            board[row][col] = 1
            solve(row + 1)
            board[row][col] = 0

solve(0)
```

OUTPUT :

```
Q . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
```

Q . . . . . .

. . . . . Q . .

. . . . . Q

. . Q . . . . .

. . . . . Q .

. . . Q . . . .

. Q . . . . . .

. . . . Q . . .

Q . . . . . .

. . . . . Q .

. . . Q . . . .

. . . . . Q . .

. . . . . Q

. Q . . . . . .

. . . . Q . . .

. . Q . . . . .